CS246: Database Management Systems Lab

Week # 08 (1 Questions, 104 Marks)
Lab session: AL1
Held on: Monday, 09-Mar-2026
Lab Timings: 14:00 to 17:00 Hours Pages: 3
Submission deadline: 17:00 hrs, 09-Mar-2026
Submissions accepted from: 16:45 hrs, 09-Mar-2026
Submissions accepted till: 18:00 hrs, 09-Mar-2026
Instructors Dr. V. Vijaya Saradhi
Every one minute delay after 17:30 hrs will result in one mark deduction
Please do not quote Network issues or technical issues after 17:30hrs
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Question 1: (104 points)

Using MySQL perform the following tasks:

Task 01 (3 Marks) Create a database named `week08`. Use this database for the following tasks.

Task 02 (56 marks) Create the following tables and load data

(a) (7 marks) A table `course` should contain 2 columns as given in the input file `courses.csv`. Choose column names of your choice.

1. (1 mark) 1st column: course id. Choose appropriate data type.
2. (1 mark) 2nd column: course title. Choose appropriate data type.

with the additional specifications:

- (1 mark) course id should be primary key
- (1 mark) Course title should not take NULL values
- (3 marks) load data from the file `courses.csv` into the table `course`

(b) (15 marks) A table `student` should contain 5 columns as given in the input file `students-credits.csv`. Choose column names of your choice. Column description is as follows:

1. (1 mark) 1st column: course id. Choose appropriate data type.
2. (1 mark) 2nd column: Roll number. Choose appropriate data type.
3. (1 mark) 3rd column: Student name. Choose appropriate data type.
4. (1 mark) 4th column: Approval status. Choose appropriate data type.
5. (1 mark) 5th column: credit status. Choose appropriate data type.

with the additional specifications:

- (2 marks) Roll number and course id together as primary key
- (3 marks) Rest of the three columns should not take NULL values
- (2 marks) course id should be a foreign key pointing to the table `course`
- (3 marks) load data from this input file.

(c) (15 marks) A table `credit` containing 5 columns. Choose column names of your choice. Column description is as detailed below:

1. (1 mark) 1st column: course id. Choose appropriate data type with required size.
2. (1 mark) 2nd column: number of lecture hours per week. Choose appropriate data type.
3. (1 mark) 3rd column: number of tutorial hours per week. Choose appropriate data type.
4. (1 mark) 4th column: number of practical hours per week. Choose appropriate data type.
5. (1 mark) 5th column: number of credits. Choose appropriate data type.

with the additional specifications:
• (1 mark) course id should be a primary key
• (4 marks) All other columns should not take NULL values
• (2 marks) course id should be a foreign key referring to course table and course id column
• (3 marks) load data from this input file.

(d) (9 marks) A table `faculty_courses` should contain two columns. Choose column names of your choice. Column description is as given below.

1. (1 mark) 1st column: course id. Choose appropriate data type.
2. (1 mark) 2nd column: faculty name. Choose appropriate data type.

with the additional specifications:
•: (2 marks) course id, faculty name together should be primary key. •: (2 marks) course id should be a foreign key pointing to the `course` table •: (3 marks) load data from this input file.

(e) (10 marks) A table `semester` should contain three columns. Choose column names of your choice. Column description is as given below.

1. (1 mark) 1st column: department name. Choose appropriate data type.
2. (1 mark) 2nd column: Semester number in roman letters. Choose appropriate data type.
3. (1 mark) 3rd column: course id. Choose appropriate data type.

with the additional specifications:
•: (2 marks) department name, semester number and course id together as primary key.
•: (2 marks) course id should be a foreign key pointing to the `course` table •: (3 marks) load data from this input file.

Task 04 (10 marks) Answer the following

(a) (5 marks) List course id and number of students auditing in each of the minor courses

(b) (5 marks) List the department name and total number of credits offered by that department Hint: extract first two characters from cid using `left()` function and use the same in the query

Task 05 (15 marks) Answer the following

- (5 marks) List the course id and number of students auditing the course in which at least four audit students are the in the course.
- (5 marks) List the course id, name of course such that the number of faculty involved in teaching that course is more than one.
- (5 marks) List the name of the faculty and number of courses that faculty is teaching such that the number of courses teaching by that faculty is more than one course.

Task 06 (10 marks) Answer the following

- (5 marks) List the course id, course name which has minimum number of credits.
- (5 marks) List the course id and faculty name who is teaching course with minimum number of credits in CSE department

Task 08 (10 marks) Answer the following

- (5 marks) Write one SQL query using the ANY operator (do not use temporary tables or multiple queries) to list the semester numbers of the BSBE department where the total credits offered in that semester are less than ANY of the semester total credits offered by the Design department.
- (5 marks) Write one SQL query using the ALL operator (do not use temporary tables or multiple queries) to list the semester numbers of the BSBE department where the total credits offered in that semester (computed as the sum of credits of all courses offered in that semester) are greater than or equal to every semester's total credits offered by the Design department.

Instructions Adhere to the following

1. Write the SQL statements corresponding to each task in a text.
2. Mobile phones are not allowed inside the lab.
3. You should upload all the SQL files in MS assignments site.